

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

EMEM gives an observation a lookup identity and a content-addressed record. Agent A passes the reference. Agent B resolves the same record and checks it before continuing.

Earth data stays in the data layer. Its reference enters the reasoning layer.

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Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

1 What survives an agent handoff?

The same observation, carried through receiver checks.

Agent A cites a source read. A relay changes the evidence. Agent B acts on it or checks the reference.

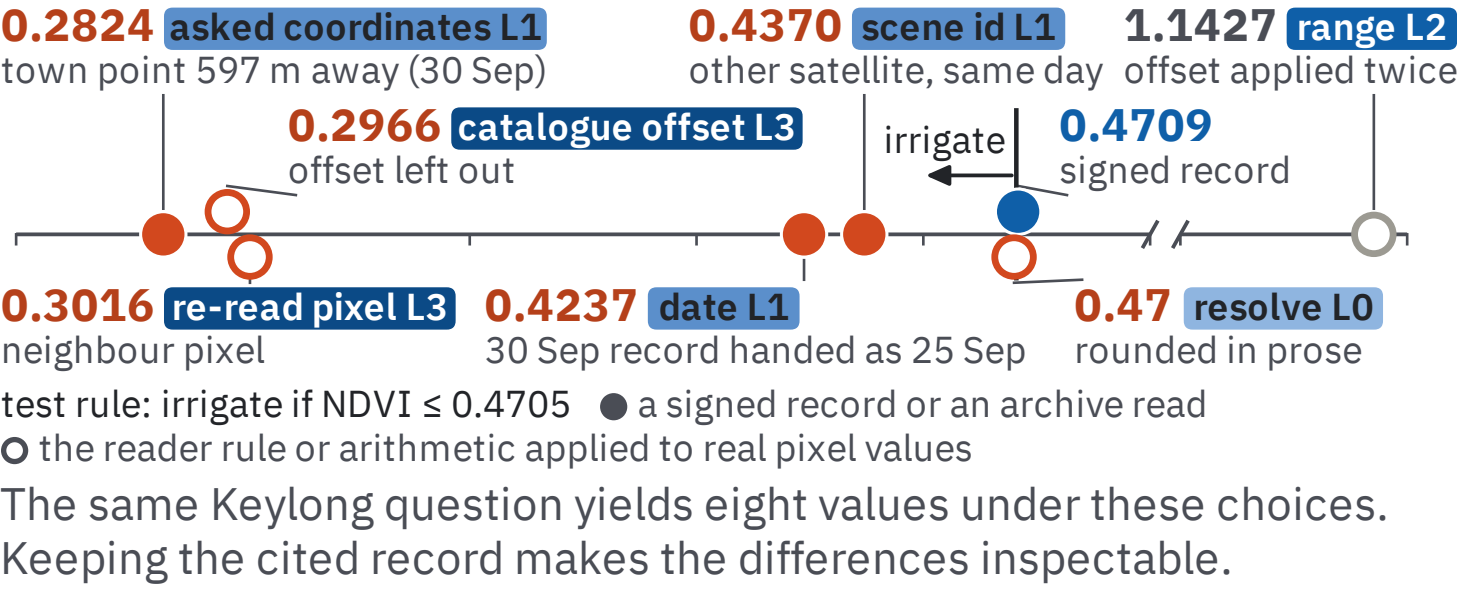


In the controlled handoff test, prose led agents to act on 254 of 276 corruptions; a reference with explicit checking instructions, 0 of 300.

2 What can change between agents?

One question, several records

Scene, pixel, offset, rounding and stale state can change the value passed on.



3 Where did the evidence change?

A changed value needs a cause

References can drift during a handoff; readouts can change between visits. Keep both questions separate.

$$\Delta Z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

World, instrument, alignment, model and noise. Δz is the later readout minus the earlier one. The attribution ledger links evidence to each term; the numeric split remains open.

IMPLEMENTED · DRIFT-ANCHOR SCORE

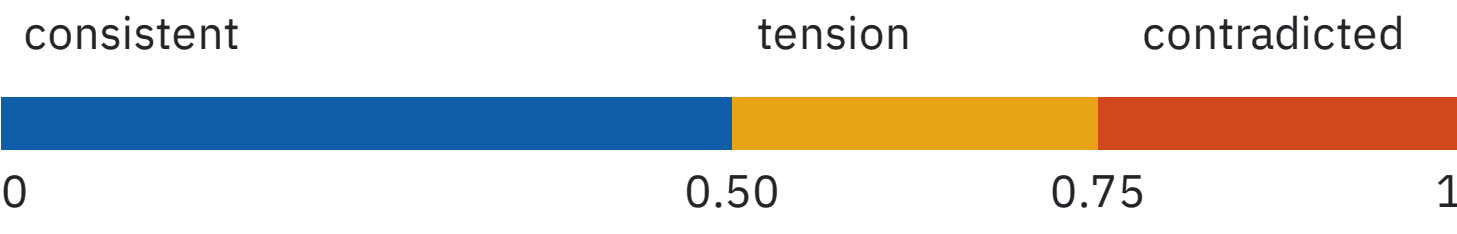
$$d = |x - a|$$

$$r = d / (3\sigma); s = r / (1 + r)$$

for finite x , a and $0 < \sigma < \infty$

Invalid σ : $s = 0$ if $d = 0$; otherwise $s = 1$.

x : output · a : anchor · σ : anchor uncertainty



Tests pin $3\sigma \rightarrow 0.50$ and $9\sigma \rightarrow 0.75$.

Score boundaries: consistent < 0.50; tension < 0.75.

The implemented anchor score measures disagreement, not its cause. SAT-042 exercises it in a reference harness; live device-to-anchor wiring remains next work.

Compute over the memory

recall · diff recover a state; compare visits

merge · trace cite a set; follow its recipe

competing · evolve keep disagreement; append

ensure · valid fill a missing read; assess age

Typed memory operations retain the evidence behind an answer.

Single-hop materialisation is implemented; multi-hop planning remains open.

20 upstream offline SDK / encoding tests passed.

Model outputs are records too

Satellite input → encoder → vector record → content address

Prithvi 1,024 values

checkpoint digest inside the record

TESSERA 128 values

product year 2024; no checkpoint digest

Archived vectors: Prithvi names a checkpoint digest; TESSERA names a product year. These encoders are retired on the deployment; stored records remain addressable.

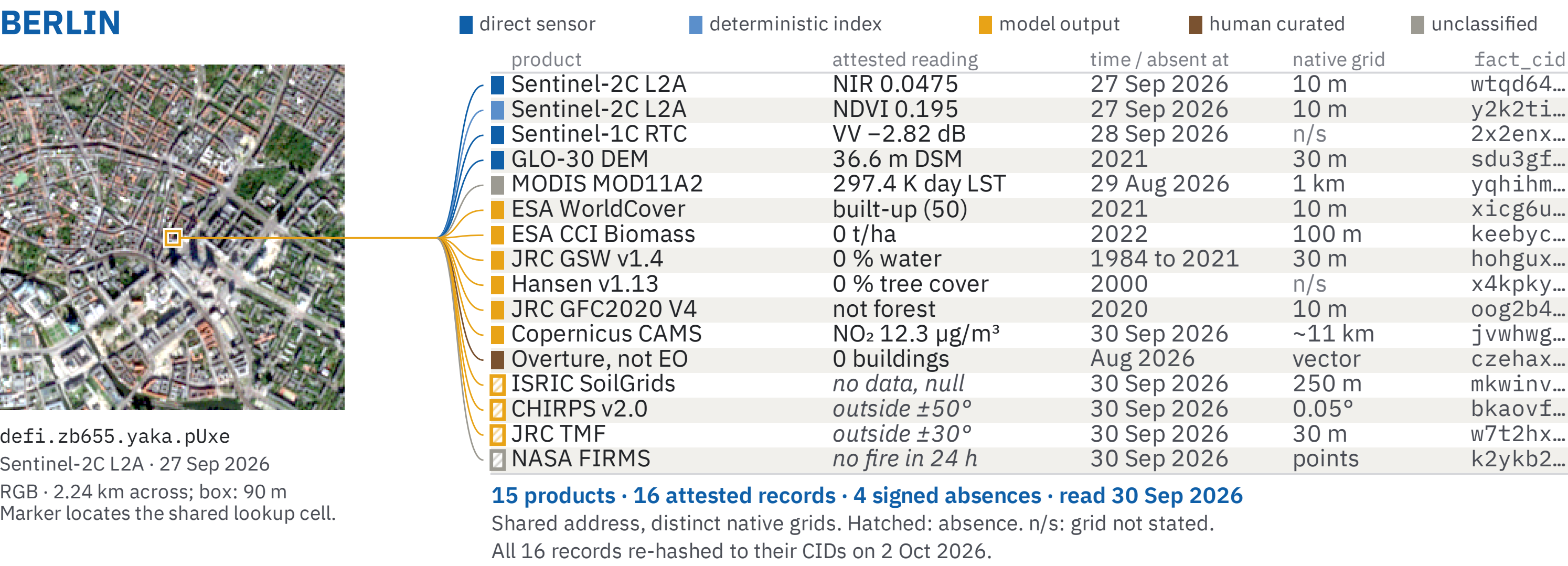
THE CONTRIBUTION
Checkable observation handoff
An agent passes a reference to a specific physical observation; the receiver resolves it and checks the evidence.
Built with standard BLAKE3, Ed25519, CBOR and Merkle logs. STAC, openEO, PROV, C2PA, RAG and temporal storage provide complementary layers.

4 What does one Berlin location reveal?

One cell, many source records

One lookup joins the evidence; each product keeps its own time, provenance and native grid.

INSPECT BERLIN Open the Keylong record and its 1,115 bytes.

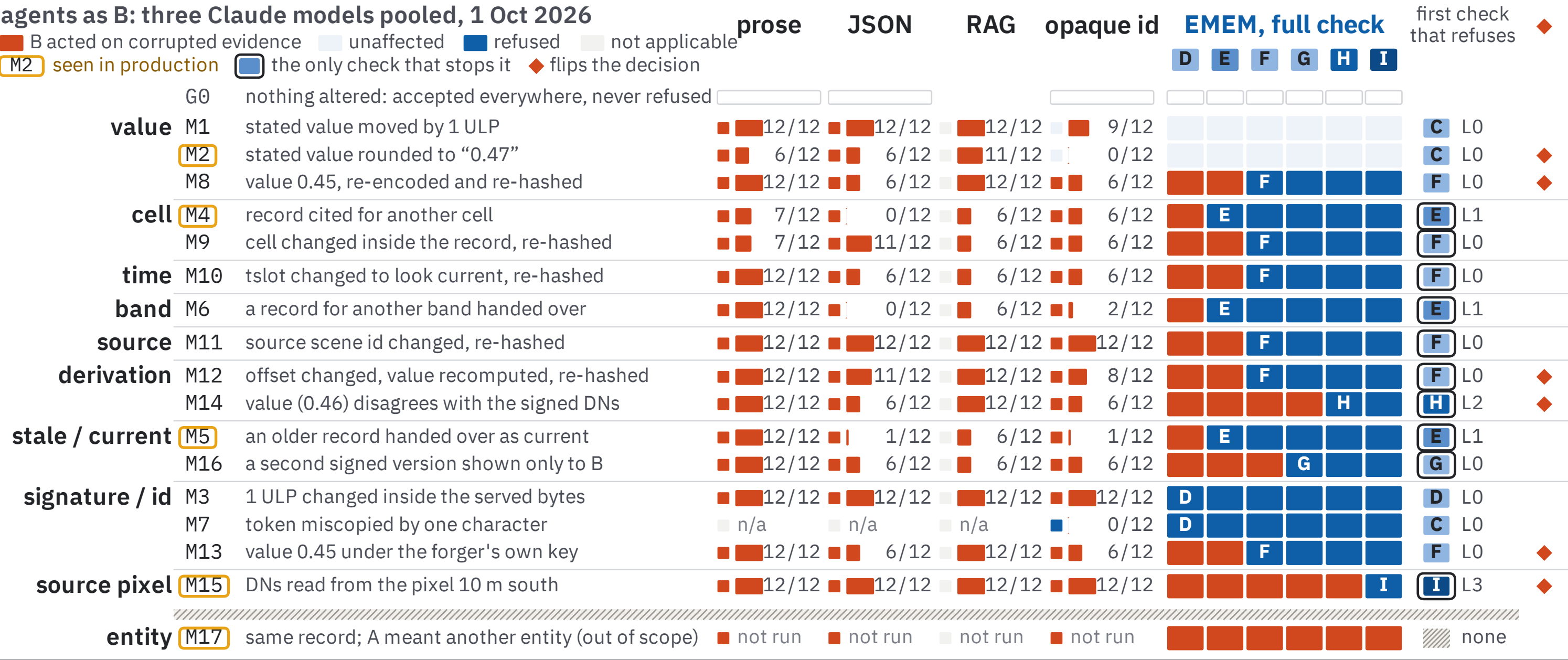


Absence is citable too: CHIRPS reports outside its latitude range. A common cell does not make these products co-registered.

5 Which changes can the receiver detect?

Checks can continue back to the source

Resolve → Re-hash → Bind → Recompute → Re-read



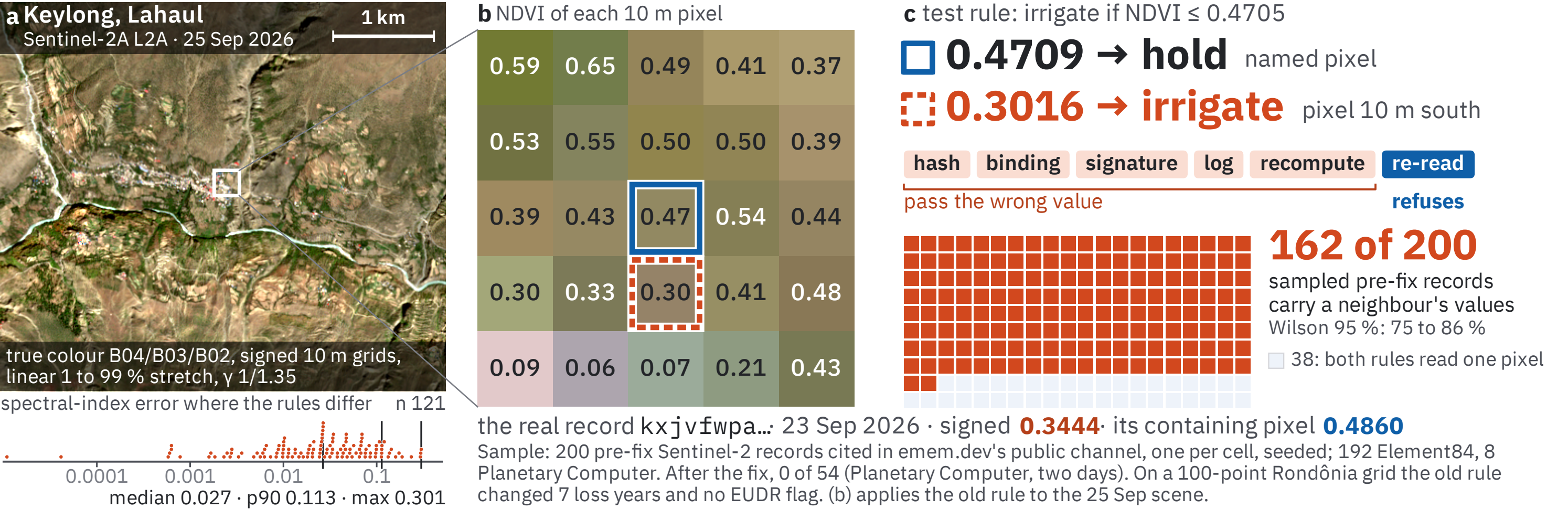
B acts on corrupted evidence

Agents' false acceptance, k of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: the deterministic ceiling. Totals pool 23 items, 1 Oct 2026. Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them. Deeper checks expose different corruptions. Checked-reference agents made the expected decision on 71 of 72 genuine controls; one was refused (pooled Claude).

Agent bars: 12 runs per item, across 23 items. JSON and an opaque id caught some changes from their fields. In the live pre-fix record case, agents acted on 4 of 4 uninstructed, 2 of 4 with resolve-and-bind instructions, and 0 of 4 with a source re-read.

6 A valid record can preserve a wrong source read

$$NDVI = (DN_8 - DN_4) / (DN_8 + DN_4 + 2o), \quad o = -1000 \text{ (product offset)}$$



11 Use emem. Carry the evidence forward.

Connect a plugin, a workflow or your own code. Public emem reads need no API key; the source is open.

Measured separately: the same record and value through 11 client paths; receipt signatures checked on 9 (30 Sep 2026).

The routes below are plugins, connectors, packages and listings. One evidence protocol, multiple agent runtimes. Setup and compatibility notes are behind CONNECT.

CONNECT Copy a token, choose your tool, run the checks. vortex-ai.github.io/esa_postexr/use/

Attested observation

→ MCP / A2A

→ Agent host / client

→ Next agent / application

ChatGPT
@emem plugin
Enable emem; paste a token.

Claude
Code plugin / MCP
Resolve from chat or terminal.

Visual Studio Code
MCP connector
Copilot agent mode: @mcp emem

Dify
Marketplace plugin
Referent Lock / EUDR workflows

Salesforce MuleSoft
Exchange listing
Open Vortex AI MCP Server

MORE CLIENTS
Cursor
Gemini CLI

PROTOCOLS
MCP · A2A
REST / OpenAPI

DEVELOPER
Python · TypeScript
Docker · LangGraph memory
GitHub source

FRAMEWORK EXAMPLES
LangChain · LlamaIndex
AutoGen · CrewAI · Mastra
Agno (see setup notes)

DISCOVERY / MIRRORS
GitHub MCP · MCP Registry
Glama · Smithery · ClawHub
Context7 · Hugging Face (older mirror)

Implementation: emem 8e9b401. Measurements: 29 Sep to 1 Oct 2026. · The linked methods connect this poster's figures to their records, source data and reproducible checks. · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066) · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)